

34 r.m.s. voltage in primary coil $V_p = \frac{180}{\sqrt{2}} = 127 \text{ V}$

In the secondary coil,

$$V_s = \frac{127}{2} = 63.6 \text{ V}$$

$$I_s = \frac{63.6}{40 + 80} = 0.530 \text{ A}$$

$$\therefore \langle P_o \rangle = I_s^2 R = (0.530^2)(80) = 22.5 \text{ W} \approx 23 \text{ W}$$

Answer: B

35 Diffraction, reflection and two source interference are evidences of the wave nature of